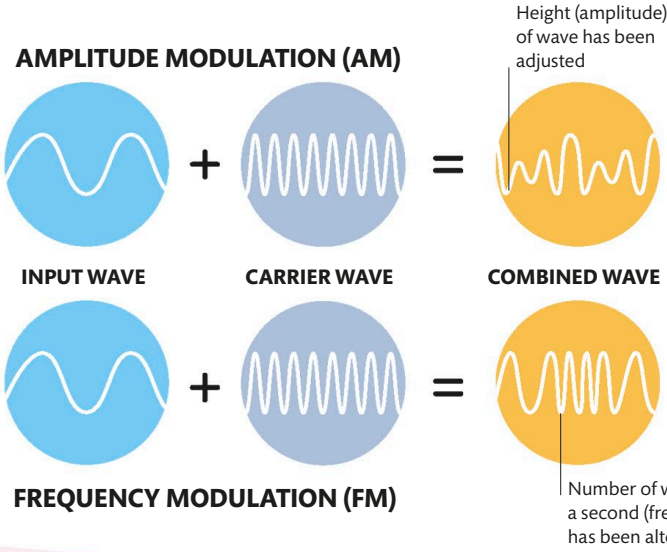
Modulation

Information is encoded into a radio wave through modulation: combining an input wave with a wave of a single frequency called a carrier wave. In AM radio broadcasting, the amplitude of the wave is altered (amplitude modulation), and in FM radio, the frequency of the wave is changed (frequency modulation). For digital radio, there are many ways to combine the input and carrier wave (see p.182).

AM and FM
AM and FM waves both look and behave differently. FM range is less than AM, but the sound quality is better and less susceptible to interference or "noise."



LIGHTNING PRODUCES VERY LOW-FREQUENCY **RADIO WAVES** CALLED **WHISTLERS**

Sky waves can cover a distance of up to 2,500 miles (4,000 km) in a single refraction from ionosphere

Sky wave
Some radio waves are refracted back to Earth's surface from the ionosphere, the charged layer of the upper atmosphere. These radio signals can travel vast distances.

SKY WAVE

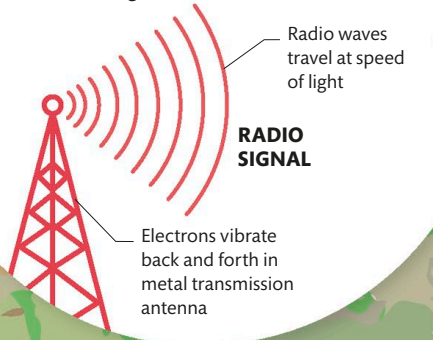
EARTH'S SURFACE

RECEIVING ANTENNA

WHAT IS LONG WAVE?

Although not defined precisely, long wave refers to wavelengths greater than 3,300 ft (1,000 m), which are usually transmitted by ground waves.

3 Transmission signal
The current travels to the transmission antenna, making electrons vibrate rapidly back and forth. This generates varying electric and magnetic fields around the antenna, radiating electromagnetic waves.



4 Radio broadcast received
The electric current passes through the radio speaker system, which causes a speaker cone to vibrate. The speaker emits sound waves, reconstructing the sound of the presenter's voice.

